Voyager 1 is about 3 kilometers (2 miles): the distance from the Empire State Building to the southern end of Greenwich Village. That's hugely less than New York to Perth.

The Earth to Saturn is even smaller: 200 meters, two east-west blocks in New York City, from the Empire State Building to Park Avenue. The Earth to Mars is just 20 meters; and the Earth to the Moon (the greatest distance humans have ever yet traveled) is just 7 centimeters—about two and a half inches!

Compare what we have achieved in going to the Moon, *two and a half inches*, with the challenge of going *halfway around the world*. That's the leap of technology required to take humans to habitable planets outside our solar system!

Travel Times with Twenty-First-Century Technology



Voyager 1 is traveling out of the solar system at 17 kilometers per second, having been boosted by gravitational slingshots around Jupiter and Saturn. In *Interstellar*, the *Endurance* travels from Earth to Saturn in two years, at an average speed of about 20 kilometers per second. The fastest speed I think rocket technology plus solar system slingshots are likely to achieve in this, the twenty-first century, is about 300 kilometers per second.

At that 300 kilometers per second, we would need 5000 years to reach Proxima Centauri and 13,000 years to reach Tau Ceti. Not a pleasant prospect!

To get there far faster in the tweny-first century, you need something like a wormhole (Chapter 14).

Far-Future Technology